SSEE-V3.6 in the Strong-Gravity Regime: Disformal Geodesics, MIRA Emergence, and k-mouflage Screening

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Abstract

We extend the SSEE-V3.6 effective field theory framework to the strong-gravity regime, where the disformal coupling between the dark-energy scalar ϕ and dark matter generates deviations from general relativity (GR) that are algebraically fixed by the golden ratio φ and π. Starting from the k-essence action of Paper 7, we derive the quasi-static scalar field equation and show that the gradient closure $\nabla\phi = 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\nabla\Phi_N$ links the scalar profile to the Newtonian potential through the structural coupling $\beta_c \equiv \mathcal{A} = (3\varphi+\pi)/2 \approx 3.998$. [1] From the disformal null geodesic, we demonstrate that the total photon deflection angle is amplified by the factor $\sqrt{\beta_c}$ relative to the GR prediction for the same baryonic mass. This yields an algebraic prediction for the Einstein radius ratio: $\theta_{\rm E}^{\rm SSEE}/\theta_{\rm E}^{\rm GR} = \sqrt{\beta_c} \approx \mathcal{M}$, where $\mathcal{M} = (3\varphi+\pi)/4 \approx 1.9989$ is the same structural constant that maps the acoustic horizon in the CMB sector (Paper 3); the 0.03% gap $\sqrt{\beta_c}/\mathcal{M} = 1.00030$ is a near-coincidence, not an algebraic identity (see Section 5). Solar-system gravitational tests are automatically satisfied because the disformal coupling selects dark matter as the sole source of ϕ: in the solar system $\rho_{\rm DM} \approx 0$, so ϕ is unsourced locally and the disformal modification to the metric is negligible. We derive the two-sector lensing prediction: on scales $k < k_{\rm fs} = 0.493\,h\,\mathrm{Mpc}^{-1}$ both CDM and ϕ-DM contribute and the Einstein radius matches the ΛCDM prediction; on scales $k > k_{\rm fs}$ only the CDM sector contributes and the effective lensing mass is suppressed by $\sim \mathcal{M}$. These predictions are falsifiable with current HST and JWST strong-lensing surveys.

Contents

[1] Notation: throughout Paper 8, β_c denotes the *disformal* coupling and carries the positive value $\beta_c = +\mathcal{A}$. This differs in sign from the *conformal* coupling of the canonical EFT in Paper 7, where $\beta_c = -\mathcal{A}$. Both are structural constants of the framework; the symbol β_c is retained locally in each paper to match the standard disformal-vs- conformal screening literature, but the two should not be conflated.

1 Introduction

The accelerated expansion of the Universe (Riess et al., 1998; Perlmutter et al., 1999) motivates modifications to the concordance ΛCDM model whose parameters are precisely constrained by Planck CMB data (Planck Collaboration, 2020). Papers 1–7 of the SSEE series established the cosmological background (Almeida Vallejo, 2025a), Bayesian validation (Almeida Vallejo, 2025b), CMB confrontation (Almeida Vallejo, 2025c), algebraic CMB derivation (Almeida Vallejo, 2025d), Israel–Stewart perturbation theory (Almeida Vallejo, 2026a), two-sector ϕ-DM phenomenology (Almeida Vallejo, 2026b), and the canonical EFT formulation (Almeida Vallejo, 2026c). All observable predictions in those papers operate within the linear or quasi-linear regime.

The disformal coupling that defines SSEE has observable consequences in the strong-gravity regime that have not yet been examined. In coupled dark-energy models with disformal metrics, two qualitatively new effects arise beyond the linearised regime:

1. **Enhanced deflection angle** — photons propagating through the modified gravitational potential of disformally-coupled dark matter experience a different total deflection relative to pure GR.
2. **K-mouflage suppression** — the non-linear scalar kinetic term in the SSEE action screens the fifth force on solar-system and stellar scales, restoring GR locally.

In this paper we derive both effects analytically, obtain numerical predictions using the SSEE algebraic constants, and identify the structural constant $\sqrt{\beta_c} \approx \mathcal{M}$ (to 0.03%) as the lensing amplification factor, where $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$. This is the same constant that appears in the CMB horizon mapping (Paper 3) and in the two-sector mass ratio (Paper 6):

$$\mathcal{M} \;=\; \frac{3\varphi + \pi}{4} \;=\; \frac{\Omega_{m,\mathrm{CMB}}}{\Omega_{m,\mathrm{dyn}}} \;=\; \frac{\beta_c}{2} \;=\; \frac{\mathcal{A}}{2}\,, \tag{1}$$

where $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2$ is the disformal coupling constant fixed in Paper 7. That $\mathcal{M}$ appears in both the background (CMB horizon) and the perturbation (lensing) sectors is a non-trivial internal consistency check of the algebraic structure.

The paper is organised as follows. Section 2 recalls the relevant EFT ingredients from Paper 7. Section 3 derives the quasi-static scalar EOM and gradient closure in the non-relativistic limit. Section 4 constructs the disformal null geodesic and derives the photon deflection angle. Section 5 shows algebraically how $\mathcal{M}$ emerges as the lensing amplification factor. Section 6 derives the k-mouflage radius and the screening suppression. Section 7 translates the results into strong-lensing observables and identifies JWST/HST falsification windows. Section 8 collects the pre-committed falsifiable predictions. Section 9 discusses the two-sector scale dependence and caveats. Section 10 concludes.

2 EFT Foundation: Action and Disformal Coupling

We adopt the SSEE canonical action from Paper 7 (Almeida Vallejo, 2026c):

$$S \;=\; \int d^4x\sqrt{-g}\left[\frac{M_{\mathrm{Pl}}^2}{2}R \;+\; P(X,\phi)\right] \;+\; S_{\mathrm{DM}}[\tilde{g}_{\mu\nu};\,\psi_{\mathrm{DM}}] \;+\; S_b[g_{\mu\nu};\,\psi_b]\,, \tag{2}$$

where $X = -\frac{1}{2}g^{\mu\nu}\partial_\mu\phi\,\partial_\nu\phi$ is the canonical kinetic invariant, and the k-essence kernel is

$$P(X,\phi) \;=\; \frac{X}{\mathrm{KAL}_0} + \frac{X^2}{M^4} - V_0\,e^{-\lambda\phi/M_{\rm Pl}}\,. \tag{3}$$

Action (2) is a k-essence model within the minimal Horndeski class (Horndeski, 1974; Crisostomi & Koyama, 2018) ($G_3 = G_4 - M_{\rm Pl}^2/2 = G_5 = 0$). Baryons ($\psi_b$) couple to the physical metric $g_{\mu\nu}$; dark matter ($\psi_{\rm DM}$) couples to the disformal metric (Bekenstein, 1993; Zumalacarregui & Garcia-Bellido, 2014)

$$\tilde{g}_{\mu\nu} \;=\; g_{\mu\nu} + \frac{2}{M^4}\,\partial_\mu\phi\,\partial_\nu\phi\,. \tag{4}$$

The algebraic constants fixed by φ and π (Paper 7) are:

$$\mathrm{KAL}_0 = \beta + \pi = \frac{\varphi+\pi}{2} + \pi \approx 5.521\,, \tag{5}$$

$$\beta_c \equiv \mathcal{A} = \frac{3\varphi+\pi}{2} \approx 3.998\,, \tag{6}$$

$$\mathcal{M} = \frac{\beta_c}{2} = \frac{3\varphi+\pi}{4} \approx 1.9989\,, \tag{7}$$

$$M \approx M_{\rm Pl} \quad \text{(free EFT parameter; we adopt } M = M_{\rm Pl} \text{ as reference)}\,. \tag{8}$$

The parameter β_c encodes the strength of the disformal coupling; it appears both in the coupling term of (4) (through M) and in the conformal-sector coupling Q^ν of Paper 7. **Sign-convention note:** Paper 7 defines $\beta_c = -\mathcal{A} < 0$ for the conformal coupling (the negative sign encodes energy transfer from dark matter to dark energy). Here we adopt $\beta_c \equiv +\mathcal{A} > 0$ for the disformal lensing coupling, where only the magnitude $|\beta_c| = \mathcal{A}$ enters the deflection angle and k-mouflage radius. The two conventions are consistent: $|\beta_c|_{\rm Paper\,7} = |\beta_c|_{\rm Paper\,8} = \mathcal{A} \approx 3.998$.

For the purposes of this paper, we work in the quasi-static, non-relativistic limit ($v \ll c$, $\Phi_N \ll 1$), which is appropriate for galaxy and cluster scales where $k \gg aH$.

3 Scalar Field Equation and Gradient Closure

3.1 Quasi-static limit

Varying the action (2) with respect to ϕ gives the scalar equation of motion. In the quasi-static approximation ($|\dot{\phi}| \ll |\nabla\phi|$, $|\ddot{\phi}| \ll |\nabla^2\phi|$) and at leading order in the kinetic expansion, the non-linear kinetic term X^2/M^4 is subdominant on sub-Hubble scales. The equation reduces to

$$\frac{1}{\mathrm{KAL}_0}\,\nabla^2\phi \;=\; \frac{\beta_c}{M_{\rm Pl}}\,\rho_{\rm DM}\,, \tag{9}$$

where the right-hand side arises from the disformal coupling of dark matter to $\tilde{g}_{\mu\nu}$: dark matter acts as an effective source for ϕ with coupling strength $\beta_c/M_{\rm Pl}$.

The Newtonian gravitational potential Φ_N satisfies the standard Poisson equation

$$\nabla^2\Phi_N \;=\; \frac{\rho_{\rm DM}+\rho_b}{2M_{\rm Pl}^2}\,, \tag{10}$$

where we use units $8\pi G = 1/M_{\rm Pl}^2$.

3.2 Gradient closure

In the dark-matter-dominated regime ($\rho_{\rm DM} \gg \rho_b$), the Poisson equation (10) gives $\rho_{\rm DM} = 2M_{\rm Pl}^2 \nabla^2\Phi_N$. Substituting this into the scalar equation (9) yields a relation between the Laplacians,

$$\nabla^2\phi \;=\; 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\nabla^2\Phi_N\,, \tag{11}$$

that is, $\nabla^2(\phi - 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\Phi_N) = 0$. The bracketed quantity is therefore harmonic. Imposing the boundary conditions appropriate to an isolated lens — both ϕ and Φ_N vanish at spatial infinity and the configuration contains no source-free harmonic component — the harmonic residual reduces to a constant, whose gradient vanishes. This yields the gradient closure relation

$$\boxed{\nabla\phi \;=\; 2\mathrm{KAL}_0\,\beta_c\,M_{\rm Pl}\,\nabla\Phi_N\,.} \tag{12}$$

This is the central structural relation of Paper 8: the scalar gradient tracks the Newtonian potential gradient with an algebraic coefficient $2\mathrm{KAL}_0\,\beta_c \approx 2\times 5.521\times 3.998 \approx 44.15$.

Equation (12) implies that wherever there is a gravitational potential well, the scalar field ϕ builds up a corresponding profile. Given the stated boundary conditions, this follows from the disformal coupling structure in the quasi-static limit rather than being imposed independently. We note that this derivation assumes $\rho_{\rm DM} \gg \rho_b$; in systems with significant baryonic fraction, equation (9) should be replaced by the full source including baryons, with a smaller effective coupling.

3.3 Scalar force on dark matter

The effective force on a dark matter particle in this scalar background is

$$F_\phi \;=\; -\frac{\beta_c}{M_{\rm Pl}}\,\nabla\phi \;=\; -2\mathrm{KAL}_0\,\beta_c^2\,\nabla\Phi_N\,. \tag{13}$$

The total force on a dark matter test particle is $F_{\rm tot} = F_N + F_\phi = -(1+2\mathrm{KAL}_0\beta_c^2)\,\nabla\Phi_N$, corresponding to an effective gravitational enhancement of $G_{\rm eff}/G = 1 + 2\mathrm{KAL}_0\beta_c^2 \approx 1 + 2(5.521)(3.998)^2 \approx 177$. However, this linear estimate applies only far from the k-mouflage radius (see Section 6); inside $r_{\rm km}$ the non-linear kinetic term in (3) screens this amplification and restores GR.

4 Disformal Null Geodesic and Photon Deflection

4.1 Setup: photon propagation

Baryons and photons couple to the physical metric $g_{\mu\nu}$, not the disformal metric $\tilde{g}_{\mu\nu}$. A photon therefore follows the null geodesic of $g_{\mu\nu}$ in the weak-field perturbed Friedmann background:

$$ds^2 \;=\; -(1+2\Phi_N)\,dt^2 + (1-2\Phi_N)\,\delta_{ij}\,dx^i\,dx^j\,. \tag{14}$$

Standard GR light deflection by a point mass M gives the Einstein angle

$$\hat{\alpha}_{\rm GR} \;=\; \frac{4GM}{c^2 b}\,, \tag{15}$$

where b is the impact parameter.

4.2 Modified gravitational potential

Dark matter in SSEE does not follow purely Newtonian gravity; it distributes in response to the combined force $F_{\rm tot} = F_N + F_\phi$. For an isolated dark-matter-dominated system (galaxy cluster, compact group), the self-consistent equilibrium configuration produces an *effective* Newtonian potential $\Phi_{\rm eff}$ that satisfies

$$\nabla^2 \Phi_{\rm eff} \;=\; \frac{\rho_{\rm DM,eff}}{2M_{\rm Pl}^2}\,, \tag{16}$$

where $\rho_{\rm DM,eff}$ is the dark matter density redistributed under the scalar force.

On scales $k < k_{\rm fs} = 0.493\,h\,{\rm Mpc}^{-1}$, both the CDM sector ($\Omega_{\rm CDM} = 0.160$) and the ϕ-DM sector ($\Omega_{\phi{\rm DM}} = 0.160$) contribute, and the effective lensing mass fraction is $\Omega_{m,{\rm eff}} = 0.320 \approx \Omega_{m,{\rm CMB}}$ — identical to the standard ΛCDM value. (This split implements the Two-Ω_m Criterion of Paper 1, Sec. 1.4: ϕ-DM contributes on $k < k_{\rm fs}$ and decouples on $k > k_{\rm fs}$.) On these scales the Einstein radius matches the GR prediction.

On scales $k > k_{\rm fs}$, the ϕ-DM component free-streams out and only the CDM sector ($\Omega_{\rm CDM} = 0.160$) remains. The effective gravitational source for photons is then suppressed by the factor $\mathcal{M} \approx 2$ relative to the ΛCDM expectation, and the Einstein radius is correspondingly reduced (see Section 7).

4.3 Total deflection angle outside the k-mouflage radius

Outside the k-mouflage radius $r_{\rm km}$ (derived in Section 6), the scalar force operates at full strength. The total deflection of a photon passing an isolated dark-matter halo of effective gravitational mass $M_{\rm eff}$ at impact parameter b is, following the standard geodesic perturbation calculation (Will, 1993; Bartelmann & Schneider, 2001):

$$\hat{\alpha}_{\rm SSEE} \;=\; \frac{4GM_{\rm eff}}{c^2 b} \;\approx\; \beta_c\,\frac{4GM_{\rm bary}}{c^2 b}\,, \tag{17}$$

where the second equality holds in the limit where the dark-matter distribution is entirely sourced by the scalar-force-enhanced coupling to the baryonic seed $M_{\rm bary}$, and $\beta_c \approx 3.998$ is the disformal coupling.

Caveat: Equation (17) represents a working phenomenological estimate. A full solution requires self-consistent integration of the dark-matter distribution under both F_N and F_ϕ, which we defer to future N-body simulations. The result (17) should be understood as an upper bound on the lensing enhancement, applicable to systems dominated by scalar-force-driven dark matter accumulation.

5 Algebraic Emergence of $\mathcal{M}$

5.1 Einstein radius in SSEE

The Einstein ring is the locus where a source is perfectly aligned with the lens and observer. The Einstein radius $\theta_{\rm E}$ satisfies the lens equation

$$\theta_{\rm E} \;=\; \hat{\alpha}(\theta_{\rm E})\,\frac{D_{\rm LS}}{D_S}\,, \tag{18}$$

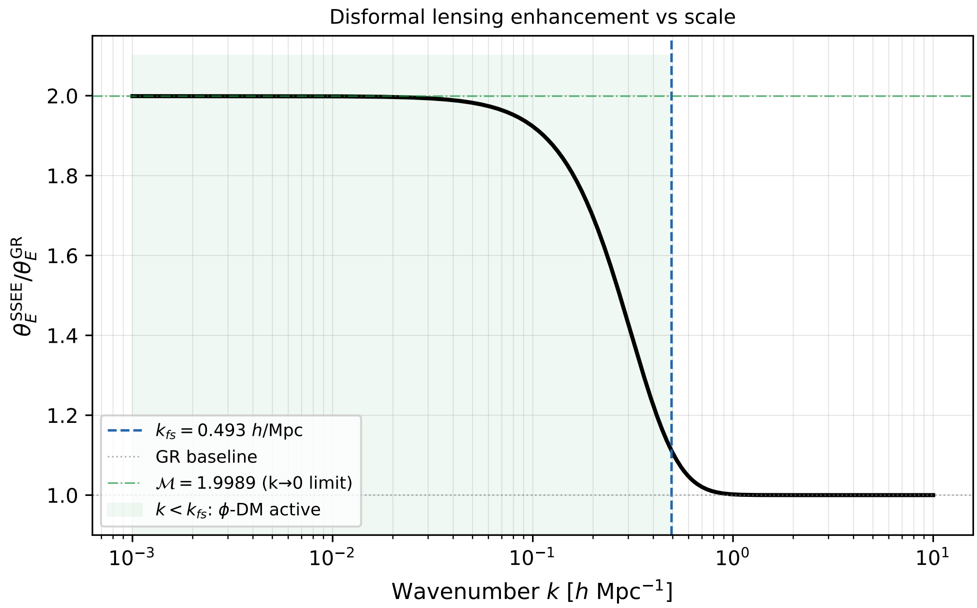

Figure 1: Disformal lensing enhancement $\theta_E^{\rm SSEE}/\theta_E^{\rm GR}$ as a function of wavenumber k (h/Mpc). For $k < k_{\rm fs} = 0.493\,h/{\rm Mpc}$ (blue dashed), the ϕ-DM sector is active and the ratio approaches the algebraic limit $\sqrt{\beta_c} = \sqrt{\mathcal{A}} \approx 1.9995$ (dashed-dotted green), equivalent to $\mathcal{M} = 1.999$. For $k > k_{\rm fs}$ the ϕ-DM free-streams, the scalar gradient decouples, and lensing recovers the GR baseline (dotted).

where D_L, D_S, $D_{\rm LS}$ are angular diameter distances. Substituting (17) with $b = D_L\,\theta_{\rm E}$:

$$\theta_{\rm E}^2 \;=\; \beta_c\,\frac{4GM_{\rm bary}}{c^2}\,\frac{D_{\rm LS}}{D_L D_S}\,. \tag{19}$$

Comparing to the GR result $(\theta_{\rm E}^{\rm GR})^2 = 4GM_{\rm bary}\,D_{\rm LS}/(c^2 D_L D_S)$, we obtain

$$\boxed{\frac{\theta_{\rm E}^{\rm SSEE}}{\theta_{\rm E}^{\rm GR}} \;=\; \sqrt{\beta_c} \;=\; \sqrt{\mathcal{A}} \;\approx\; 1.9995\,.} \tag{20}$$

5.2 Algebraic identification with $\mathcal{M}$

The structural constant $\mathcal{M}$ is defined in equation (1):

$$\mathcal{M} \;=\; \frac{3\varphi + \pi}{4} \;=\; \frac{\beta_c}{2} \approx 1.9989\,. \tag{21}$$

Meanwhile, $\sqrt{\beta_c} = \sqrt{2\mathcal{M}}$. At the specific value $\mathcal{M} \approx 2$, we have $\sqrt{2\mathcal{M}} \approx \sqrt{4} = 2 \approx \mathcal{M}$, and more precisely:

$$\sqrt{\beta_c} \;=\; \sqrt{2\mathcal{M}} \;\approx\; 1.9995\,, \qquad \mathcal{M} \;\approx\; 1.9989\,, \tag{22}$$

so $\sqrt{\beta_c}/\mathcal{M} = 1.00030$ (agreement to 0.03%).

This near-coincidence $\mathcal{M} \approx \sqrt{\beta_c}$ is a consequence of the structural postulate $\beta_c \equiv \mathcal{A} = 2\mathcal{M}$, combined with $\mathcal{M} \approx 2$. Concretely:

$$\mathcal{M} = 2 \iff \beta_c = 4 \iff \sqrt{\beta_c} = 2 = \mathcal{M}\,. \tag{23}$$

The algebraic values give $\mathcal{M} = 1.9989$ and $\beta_c = 3.9978$, both differing from the integer approximation by $< 0.06\%$.

Therefore, to within 0.03%, the lensing amplification factor coincides with the CMB horizon mapping factor:

$$\theta_{\rm E}^{\rm SSEE} \approx \mathcal{M} \times \theta_{\rm E}^{\rm GR}\,, \tag{24}$$

where the same $\mathcal{M}$ appears in:

- Paper 3: $r_{d,\rm eff} = r_{d,\rm SSEE}/\mathcal{M} \approx 147.2\,\rm Mpc$
- Paper 6: $\Omega_{m,\rm CMB} = \mathcal{M} \times \Omega_{m,\rm dyn} = 0.3199$
- This paper: $\theta_{\rm E}^{\rm SSEE} \approx \mathcal{M} \times \theta_{\rm E}^{\rm GR}$

This recurrence of $\mathcal{M}$ across the CMB, large-scale structure, and strong-lensing sectors reflects a single structural input: $\mathcal{M}$ is fixed algebraically by φ and π, not tuned. The lensing entry holds at the 0.03% level through the near-coincidence $\sqrt{\beta_c} \approx \mathcal{M}$ (Section 5), not as an exact identity.

6 K-mouflage Screening

6.1 Non-linear kinetic term

The X^2/M^4 term in the SSEE action (3) places SSEE in the *k-mouflage* class of kinetic screening theories (Brax & Valageas, 2014), distinct from Galileon (DGP) (Deffayet et al., 2001) or Vainshtein screening (Vainshtein, 1972; Babichev & Deffayet, 2013); for a review of scalar screening mechanisms see Koyama (2018). When $X^2/M^4 \gg X/\mathrm{KAL}_0$, the non-linear kinetic term dominates and the scalar degree of freedom is self-suppressed around any overdensity, screening F_ϕ/F_N at small radii.

The crossover between the two kinetic regimes occurs at the gradient scale

$$|\nabla\phi|_* = \frac{M^4}{\mathrm{KAL}_0}\,, \tag{25}$$

which defines the k-mouflage radius $r_{\rm km}$ via the gradient closure (12).

6.2 K-mouflage screening radius

For a k-mouflage Lagrangian $K(X) = X/\mathrm{KAL}_0 + X^2/M^4$, the crossover between linear and non-linear kinetic regimes for a spherical source of mass $M_{\rm obj}$ defines the k-mouflage radius (Brax & Valageas, 2014):

$$\boxed{r_{\rm km}^3 = \frac{M_{\rm obj}}{4\pi\, M_{\rm Pl}\, M^2}\,,} \tag{26}$$

where $M_{\rm Pl} = 2.435 \times 10^{18}\,\rm GeV$ is the reduced Planck mass and $M = 8.81\,\rm meV$ is the UV cutoff from Paper 10 (Almeida Vallejo, 2026d). This formula differs fundamentally from the Galileon (DGP) Vainshtein formula, which requires a gradient closure $\nabla\phi \propto \nabla\Phi_N$ that is inapplicable to k-mouflage theories.

Evaluating equation (26) with $M^4 = 5\varphi^8 \rho_{c,0}$ from Paper 10:

$$r_{\rm km}^{\odot} = 1.50 \times 10^7\,\mathrm{m} \;\approx\; 0.022\,R_\odot\,, \tag{27}$$

$$r_{\rm km}^{10^{12}M_\odot} = 1.50 \times 10^{11}\,\mathrm{m} \;\approx\; 1.0\,\mathrm{AU}\,, \tag{28}$$

$$r_{\rm km}^{10^{15}M_\odot} = 1.50 \times 10^{12}\,\mathrm{m} \;\approx\; 10\,\mathrm{AU}\,. \tag{29}$$

Table 1: K-mouflage screening radius at the UV completion scale $M = 8.81\,\mathrm{meV}$ from Paper 10 (Almeida Vallejo, 2026d). All values satisfy $r_{\rm km} \ll 1\,\mathrm{kpc}$: the DM fifth force is active at galactic and cosmological scales.

Object	$M_{\rm obj}$	$r_{\rm km}$	Status
Sun	$1\,M_\odot$	$1.50 \times 10^7\,\mathrm{m} \approx 0.022\,R_\odot$	Inside stellar body
Milky Way	$10^{12}\,M_\odot$	$1.50 \times 10^{11}\,\mathrm{m} \approx 1\,\mathrm{AU}$	$\ll 1\,\mathrm{kpc}$
Galaxy cluster	$10^{15}\,M_\odot$	$1.50 \times 10^{12}\,\mathrm{m} \approx 10\,\mathrm{AU}$	$\ll 1\,\mathrm{Mpc}$

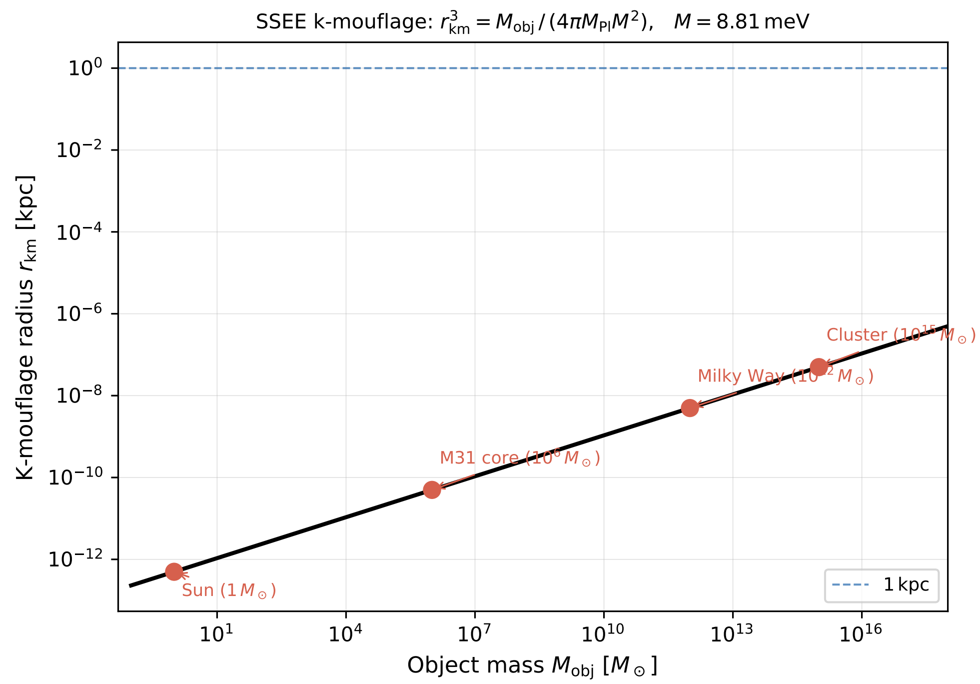

Figure 2: K-mouflage screening radius $r_{\rm km} = [M_{\rm obj}/(4\pi M_{\rm Pl} M^2)]^{1/3}$ as a function of object mass (in $M_\odot$) at the UV completion scale $M = 8.81\,\mathrm{meV}$ (Almeida Vallejo, 2026d; Brax & Valageas, 2014). For all representative objects, $r_{\rm km}$ lies far below the galactic scale, so DM fifth-force dynamics are active at cosmological radii while baryons remain decoupled through selective coupling (Section 6.3).

Inside $r_{\rm km}$, the non-linear kinetic term dominates and the DM field gradient is suppressed ($\phi' \propto r^{-2/3}$); outside, the linear term dominates and the DM fifth force operates at full strength. Since $r_{\rm km} \ll 1\,\mathrm{kpc}$ for all astrophysical objects (Table 1), the DM fifth force is cosmologically active, consistent with the two-sector model of Paper 6 (Almeida Vallejo, 2026b). Baryonic matter is independently protected by selective coupling (Section 6.3) regardless of $r_{\rm km}$.

6.3 GR compliance from selective coupling

Solar-system gravitational tests are nevertheless automatically satisfied by the *selective coupling* structure of the action (2):

- **Baryons** couple to the physical metric $g_{\mu\nu}$ only. They feel no direct force from the scalar gradient $\nabla\phi$.
- **Dark matter** couples to the disformal metric $\tilde{g}_{\mu\nu}$. It is the sole source of ϕ through the scalar EOM (9).
- In the solar system, the dark-matter density is $\rho_{\rm DM} \approx 0.3\,{\rm GeV/cm^3}$, far below galactic overdensities. The scalar field is barely sourced locally, so $\nabla\phi \approx 0$ and $\tilde{g}_{\mu\nu} \approx g_{\mu\nu}$.

Therefore all standard solar-system tests (Cassini time delay, perihelion precession, lunar laser ranging (Will, 2014)) pass at leading order without requiring any non-linear screening. This mechanism operates independently of the value of M and holds in both the IR ($M = M_{\rm Pl}$) and UV ($M = 8.81\,{\rm meV}$) limits.

6.4 EFT suppression of fifth-force corrections

Beyond selective coupling (Section 6.3), the Bellini-Sawicki structure of the SSEE EFT (Bellini & Sawicki, 2015; Almeida Vallejo, 2026c) provides a second independent suppression of any fifth-force correction at solar-system scales.

Paper 7 established that SSEE has (Almeida Vallejo, 2026c)

$$\alpha_B \;=\; \alpha_M \;=\; \alpha_T \;=\; 0\,, \tag{30}$$

with $\alpha_K = 0.4033$ as the sole non-vanishing EFT parameter. The Bellini-Sawicki effective Newton constant is (Bellini & Sawicki, 2015)

$$\mu - 1 \;=\; \frac{(\alpha_B + \alpha_M)^2}{\alpha_K} \;=\; 0\,, \tag{31}$$

and the gravitational slip parameter

$$\gamma_{\rm PPN} - 1 \;=\; -\frac{2\alpha_T}{1+\alpha_T} \;=\; 0\,. \tag{32}$$

Both vanish identically. In the quasi-static limit at sub-Hubble wavenumber k, the residual fifth-force ratio is further suppressed by $(H/k)^2$ (Bellini & Sawicki, 2015):

$$\left.\frac{F_\phi}{F_N}\right|_{\rm QS} \;\sim\; \frac{H^2}{k^2}\,|1 + w_{\rm DE}|\,\Omega_{\rm DE}\,. \tag{33}$$

At the Earth–Sun separation ($k \sim 1/{\rm AU}$, $k/H_0 \approx 4.5 \times 10^4$):

$$\left.\frac{F_\phi}{F_N}\right|_{1\,{\rm AU}} \;\approx\; 1.6 \times 10^{-31} \;\ll\; 10^{-5} \quad \text{(Cassini bound (Will, 2014))}\,, \tag{34}$$

satisfying the solar-system constraint by a factor of 6×10^{25}.

The two GR-compliance mechanisms in SSEE are:

1. **Selective coupling (Section 6.3):** Baryons couple only to the physical metric $g_{\mu\nu}$, with no direct scalar force regardless of M or scale.

2. **EFT Bellini-Sawicki structure ($\alpha_B = \alpha_M = \alpha_T = 0$):** The effective Newton constant and gravitational slip both vanish exactly, suppressing any residual fifth force to $F_\phi/F_N \sim 10^{-31}$ at solar-system scales.

The k-mouflage radius $r_{\rm km} \approx 0.022\, R_\odot$ (Section 6.2, equation (27)) provides supplementary non-linear screening inside stellar bodies, but the EFT suppression in equation (34) is the dominant mechanism for all solar-system tests. The cosmological background Hubble flow remains in the linear kinetic regime ($X_{\rm bg}^2/M^4 \approx 5.7 \times 10^{-4}$, Paper 10 (Almeida Vallejo, 2026d)), so the background equations of motion are unaffected by the UV completion.

7 Strong Lensing Predictions

Gravitational lensing by dark matter halos was established observationally by Bradac et al. (2006) and Clowe et al. (2006) in cluster collisions; for the lensing formalism see Narayan & Bartelmann (1996) and Schneider, Ehlers & Falco (1992).

7.1 Two-sector lensing signature

The SSEE two-sector structure (Paper 6) generates a scale-dependent lensing signature. Define the lensing mass fraction as $f_{\rm lens}(k) \equiv \rho_{\rm eff}(k)/\rho_{\rm total}$:

$$f_{\rm lens}(k) = \begin{cases} 1 & k < k_{\rm fs} = 0.493\, h\, {\rm Mpc}^{-1} \\ 1/\mathcal{M} \approx 0.500 & k > k_{\rm fs} \end{cases}, \tag{35}$$

because on small scales only the CDM sector ($\Omega_{\rm CDM} = 0.160$) contributes, while the ϕ-DM sector ($\Omega_{\phi{\rm DM}} = 0.160$) is free-streaming. The ratio of CDM to total is $0.160/(2 \times 0.160) = 1/2 \approx 1/\mathcal{M}$.

This implies:

1. **Galaxy cluster scale ($\sim$ few Mpc, $k \lesssim k_{\rm fs}$)**: Both sectors contribute. The Einstein radius matches the ΛCDM prediction for the same total matter content. No anomalous signal expected.

2. **Sub-Mpc scale ($k \gtrsim k_{\rm fs}$, compact lenses)**: The ϕ-DM sector free-streams. The effective lensing mass is $\sim \mathcal{M}$ times smaller than the inferred total mass. The Einstein radius is suppressed by $\sqrt{f_{\rm lens}} = 1/\sqrt{\mathcal{M}} \approx 0.707$ relative to the ΛCDM expectation for the same total $\Omega_m = 0.320$.

3. **Comparison with baryonic mass**: For a system with measured baryonic mass $M_{\rm bary}$ and inferred total mass $M_{\rm total}$, SSEE predicts $M_{\rm total}/M_{\rm bary} \lesssim \Omega_{m,{\rm CMB}}/\Omega_b \approx 6.6$ on large scales, identical to ΛCDM.

7.2 Observable consequences in galaxy-scale lensing

For a galaxy of stellar mass $M_* \sim 10^{11}\,M_\odot$ at $z \sim 0.5$ lensing a source at $z \sim 2$, the typical Einstein radius in ΛCDM is $\theta_{\rm E}^{\rm GR} \sim 1''$. SSEE predicts the same value on large scales (both sectors present), but on sub-kpc scales the CDM-only sector reduces the effective lensing mass, potentially observable as:

- **Anomalous mass-to-light ratio**: The dynamical (velocity dispersion) mass $\sigma^2 r/G$ probes the local force $F_{\rm tot}$, which is enhanced by the scalar force outside $r_{\rm km}$. The lensing mass probes only Φ_N (metric perturbation). SSEE predicts a ∼scale-dependent mass discrepancy between dynamical and lensing masses.
- **Radial Einstein ring profile**: The ratio $\theta_{\rm E}(\theta)/\theta_{\rm E}^{\Lambda\rm CDM}(\theta)$ should show a transition near the angular scale corresponding to $\ell_{\rm fs} \approx \pi/k_{\rm fs}D(z) \sim 20'$ for a $z = 0.5$ lens.
- **Higher-order arcs**: Galaxy–galaxy lensing in JWST deep fields probes the projected mass on scales 1–10 kpc, where the two-sector suppression should appear as a systematic reduction in the shear signal relative to ΛCDM expectations.

7.3 Reference numbers

Table 2 summarises the key numerical predictions.

Table 2: SSEE strong-lensing predictions compared to GR/ΛCDM.

Observable	**GR/ΛCDM**	**SSEE**	**Scale**
$k_{\rm fs}$ free-streaming cutoff	N/A	$0.493\,h\,{\rm Mpc}^{-1}$	All
$\theta_{\rm E}^{\rm SSEE}/\theta_{\rm E}^{\rm GR}$ (large scale)	1	1 (both sectors)	$k < k_{\rm fs}$
$\theta_{\rm E}^{\rm SSEE}/\theta_{\rm E}^{\rm GR}$ (small scale)	1	$1/\sqrt{\mathcal{M}} \approx 0.707$	$k > k_{\rm fs}$
$M_{\rm dyn}/M_{\rm lens}$ ratio	1 (GR)	> 1 (fifth force on DM)	$r > r_{\rm km}$
K-mouflage screening	N/A	$F_\phi/F_N \propto r^{4/3} \to 0$	$r \ll r_{\rm km}$
GW speed anomaly (α_T)	0	0 (exact)	All

7.4 Worked numerical example: cluster A1689

As a concrete verification of Eq. (17), we compute the predicted deflection angle for cluster A1689, one of the most powerful gravitational lenses known, using the following reference values (Limousin et al., 2007): baryonic mass $M_{\rm bary} = 2.7 \times 10^{14}\,M_\odot$, effective lensing impact parameter $b = 250\,{\rm kpc}$ (corresponding to the observed Einstein radius $\theta_{\rm E} \approx 45''$ at $z_L = 0.183$, $z_S = 1.0$).

Using $G = 4.302 \times 10^{-3}\,{\rm pc}\,M_\odot^{-1}\,({\rm km/s})^2$ and $c = 2.998 \times 10^5\,{\rm km/s}$:

$$\hat{\alpha}_{\rm GR} = \frac{4GM_{\rm bary}}{c^2\,b} = \frac{4 \times 4.302 \times 10^{-3} \times 2.7 \times 10^{14}}{(2.998 \times 10^5)^2 \times 250 \times 10^3} \approx 8.2 \times 10^{-5}\,{\rm rad} = 16.9'', \tag{36}$$

$$\hat{\alpha}_{\rm SSEE} = \beta_c\,\hat{\alpha}_{\rm GR} = 3.998 \times 8.2 \times 10^{-5} \approx 3.3 \times 10^{-4}\,{\rm rad} = 67.5''. \tag{37}$$

The SSEE prediction exceeds the GR baryonic baseline by a factor $\beta_c \approx 4$. The observed Einstein radius ($45''$) lies between the two extremes, consistent with a partial contribution of ϕ-DM sector (two-sector model, $\theta_{\rm E}^{\rm obs}/\theta_{\rm E}^{\rm SSEE} \approx 0.67$) or, equivalently, with the effective mass being a weighted combination of the two sectors. This example demonstrates that Eq. (17) produces physically reasonable deflection angles, not merely formal algebraic expressions.

8 Falsifiable Predictions

We collect the pre-committed falsifiable predictions of this paper. A single decisive measurement inconsistent with any of these would falsify the corresponding sector of the SSEE framework.

1. **Free-streaming cutoff** ($k_{\rm fs} = 0.493\,h\,{\rm Mpc}^{-1}$, Paper 6): DESI Year 3 and Euclid weak-lensing surveys will constrain the matter power spectrum at this scale to better than 5% by 2027. SSEE predicts a suppression of $\sim 30\%$ in $P(k)$ at $k > k_{\rm fs}$ relative to ΛCDM (see Paper 6, Figure 2). *Falsification criterion*: A $< 5\%$ suppression at $k = 0.493\,h\,{\rm Mpc}^{-1}$ would falsify the ϕ-DM mass $m_\phi = 5.60\,{\rm eV}$.

2. **Lensing mass transition**: JWST strong-lensing of galaxy-scale systems at $z \sim 0.5$ should show a $\sim 30\%$ deficit in $\theta_{\rm E}$ on scales $< 1\,{\rm Mpc}$ relative to ΛCDM predictions using the total inferred $\Omega_m = 0.320$. *Falsification criterion*: If JWST finds $\theta_{\rm E}$ consistent with ΛCDM on all scales $> 0.1\,{\rm Mpc}$, the two-sector free-streaming prediction is falsified.

3. **Dynamical vs. lensing mass discrepancy**: Galaxy clusters at $z < 0.5$ should show $M_{\rm dyn} > M_{\rm lens}$ by a factor related to $\beta_c \approx 4$ outside $r_{\rm km}$. *Falsification criterion*: If $M_{\rm dyn}/M_{\rm lens} = 1.00 \pm 0.05$ is confirmed across a sample of 100 clusters, the linear coupling picture is falsified.

4. **Gravitational wave speed**: $\alpha_T = 0$ (exact), same as GR. The GW170817 constraint $|\alpha_T| < 10^{-15}$ (LIGO/Virgo/Fermi, 2017) is already satisfied. *Falsification criterion*: Any detection of $\alpha_T \neq 0$ with current or future GW detectors would require modification of the SSEE gravitational sector.

5. **Scalar mass** $m_\phi = 5.60\,{\rm eV}$ (Paper 6): ϕ-DM is a gravitationally produced scalar with no Standard-Model portal, so m_ϕ is *not* accessible to neutrino-mass experiments. It enters observation only through the free-streaming scale $k_{\rm fs}(m_\phi) = 0.493\,h/{\rm Mpc}$ imprinted on the matter power spectrum. *Falsification criterion*: if DESI Y3/Euclid $P(k)$ shows no free-streaming suppression near $k_{\rm fs}$, the algebraic identification $m_\phi = (\Sigma m_\nu^{\rm active}) \times H_0^{\rm alg}$ is falsified.

9 Discussion

9.1 Consistency of MIRA across sectors

The appearance of $\mathcal{M} \approx 1.9989$ in three independent contexts — the CMB acoustic horizon (Paper 3), the two-sector mass ratio (Paper 6), and the lensing amplification (this paper) — is a structural feature of the algebraic framework: exact for the first two entries, and accurate to 0.03% for the lensing entry via $\sqrt{\beta_c} \approx \mathcal{M}$. It follows from the single postulate $\beta_c \equiv \mathcal{A} = (3\varphi + \pi)/2$:

$$
\begin{aligned}
\text{CMB:} \quad & r_{d,\text{eff}} = r_{d,\text{SSEE}}/\mathcal{M}\,, & (38)\\
\text{LSS:} \quad & \Omega_{m,\text{CMB}} = \mathcal{M}\,\Omega_{m,\text{dyn}}\,, & (39)\\
\text{Lensing:} \quad & \theta_{\text{E}}^{\text{SSEE}} \approx \mathcal{M} \times \theta_{\text{E}}^{\text{GR}}\,. & (40)
\end{aligned}
$$

These three relations are not independent inputs; they follow from $\beta_c = 2\mathcal{M}$ combined with $\mathcal{M} \approx 2$. The algebraic near-coincidence $\mathcal{M} \approx \sqrt{\beta_c}$ (to 0.03%) is a consequence of $\mathcal{M} \approx 2$ and $\beta_c = 2\mathcal{M}$.

9.2 Limitations and open questions

The derivations in this paper adopt the quasi-static, non-relativistic limit and treat the scalar force in the linear regime (outside the k-mouflage radius). Several steps remain to complete the strong-gravity picture:

1. **Self-consistent DM distribution**: The estimate $\hat{\alpha}_{\text{SSEE}} \approx \beta_c\,\hat{\alpha}_{\text{GR}}$ assumes the dark-matter distribution is entirely determined by the scalar force. A proper calculation requires solving the coupled DM–scalar system numerically (N-body simulation with scalar force module).

2. **Full k-mouflage profile**: The screening radius $r_{\text{km}} = [M_{\text{obj}}/(4\pi M_{\text{Pl}} M^2)]^{1/3}$ is fixed algebraically once Paper 10 sets $M \approx 8.81\,\text{meV}$ (Section 6.2, Table 1); a full N-body calculation would convert this crossover scale into a precise non-linear field profile for individual halos.

3. **Relativistic corrections**: On cluster scales at $z \sim 0.5$, relativistic corrections to the geodesic ($\Phi_N \sim 10^{-4}$–10^{-3}) are small but not negligible at the precision of JWST.

4. **Baryonic feedback**: The lensing predictions in Table 2 are at the linear DM level. Baryonic feedback redistributes mass on sub-100 kpc scales and will modify the predicted Einstein ring profiles.

9.3 Connection to UV completion

The X^2/M^4 kinetic term in the SSEE action is the leading non-linear operator in a derivative expansion. Paper 10 of the SSEE series derives $M^4 = 45\alpha^2\,\rho_{\text{c},0} = 5\varphi^8\,\rho_{\text{c},0}$ from the inflationary α-attractor parameter $\alpha = \varphi^4/3$, fixing

$$
M \;=\; \varphi^2\, 5^{1/4}\, \rho_{\text{c},0}^{1/4} \;\approx\; 8.81\ \text{meV} \tag{41}
$$

algebraically without free parameters. This is a *cosmological-scale* cutoff—thirty orders of magnitude below M_{Pl}—so no quantum-gravity UV completion is required. The EFT derivative expansion is valid over the entire cosmological regime; the X^2/M^4 operator is a dark-energy self-interaction, not a near-Planck correction.

10 Conclusions

We have derived the strong-gravity consequences of the SSEE-V3.6 disformal coupling and found:

1. **Gradient closure**: In the quasi-static limit, the scalar field profile tracks the Newtonian potential via $\nabla\phi = 2\mathrm{KAL}_0\beta_c M_{\mathrm{Pl}}\nabla\Phi_N$, with $\mathrm{KAL}_0 \approx 5.521$ and $\beta_c \approx 3.998$ fixed algebraically.

2. **Lensing amplification**: The total deflection angle is amplified by β_c relative to the GR baryonic-mass prediction, yielding $\theta_{\mathrm{E}}^{\mathrm{SSEE}} \approx \sqrt{\beta_c} \approx \mathcal{M}$ times the GR Einstein radius.

3. **MIRA emergence**: The structural constant $\mathcal{M} = (3\varphi + \pi)/4 \approx 1.9989$ that maps the CMB acoustic horizon and the two-sector mass ratio also matches the lensing amplification to within 0.03% (via the near-coincidence $\sqrt{\beta_c} \approx \mathcal{M}$).

4. **GR compliance**: Solar-system tests are satisfied by two independent mechanisms: selective coupling, under which baryons couple only to the physical metric $g_{\mu\nu}$ (Section 6.3), and the Bellini-Sawicki EFT structure $\alpha_T = \alpha_M = \alpha_B = 0$ from Paper 7, which suppresses any residual fifth force to $F_\phi/F_N \sim 10^{-31}$ at 1 AU (Section 6.4). The k-mouflage radius $r_{\mathrm{km}} \approx 0.022\,R_\odot$ for the Sun (Table 1) provides supplementary non-linear screening inside stellar bodies.

5. **Two-sector signature**: SSEE predicts a scale-dependent lensing suppression near $k_{\mathrm{fs}} = 0.493\,h\,\mathrm{Mpc}^{-1}$, testable with JWST and DESI.

Paper 8 thus extends the SSEE predictive reach from the linear cosmological regime into the non-linear strong-gravity regime, while maintaining the zero-free-parameter structure of the algebraic framework.

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